

Backend Roadmap

PYTHON

- Introduction to Python
- Python Installation
- Syntax Compared to Other Languages
- Variables and Data types
- Operators (Arithmetic , Assignment , Comparison)
- Operators (Logical , Bitwise , Identity , Membership)
- Sequence Types (List, Tuple)
- Mapping Type (Dictionary)
- Set Types (Set)
- Conditional Statements (if , elif , else)
- Loop Statements (for , while)
- Control Statements
- Pattern Programs
- Functions , parameters , arguments
- Lambda Function, Decorators
- OOP Concepts (Class and Object)
- Inheritance , Polymorphism
- Encapsulation , Data Abstraction
- Exception Handling (Try, Except, Else, Finally)
- Threading Essentials
- Modules and JSON
- File Handling

SQL

- What is SQL and Database ? RDBMS vs Non-RDBMS
- Database Installation and Set up
- Tables, Rows (Tuples), Columns (Attributes)
- Data Types : INT, VARCHAR, DATE, BOOLEAN, DECIMAL
- Constraints: PK, FK, UQ, CHK, NN, DF

- DDL : CREATE , ALTER , DROP , TRUNCATE
- DML : INSERT , UPDATE , DELETE
- DQL : SELECT , Filtering , Sorting , Limiting , Grouping
- Operators : Arithmetic , Comparisonal , Logical , Special
- Clauses : ORDER BY , GROUP BY , HAVING , WHERE
- Aggregation : COUNT , MAX , MIN ,AVG , SUM
- Function : String , Date Time , Math
- Joins : INNER , LEFT , RIGHT , FULL OUTER , SELF
- TCL : START TRANSACTION, COMMIT , ROLLBACK
- DCL : GRANT , REVOKE
- Views : Simplified Queries
- Index : Faster Search
- Procedures, Triggers

DJANGO

- What is Django Framework ?
- Installation and Setup
- Creating a Project and an Application
- Folder Structure
- Requests and Responses
- MVT (Model , Views , Template)
- Django Template Language
- Database , Migrations
- Admin Page, Superuser
- Static Files and Forms
- Generic Views
- Settings File Update
- Admin Customization
- API and Security
- REST API Overview
- Login-Signup Page , Accounts and Auth
- Python to SQL Connection
- Search Options

- Final Project
- Doubt Clarification
- Django Project Presentation
- Final Mock Interview + Support